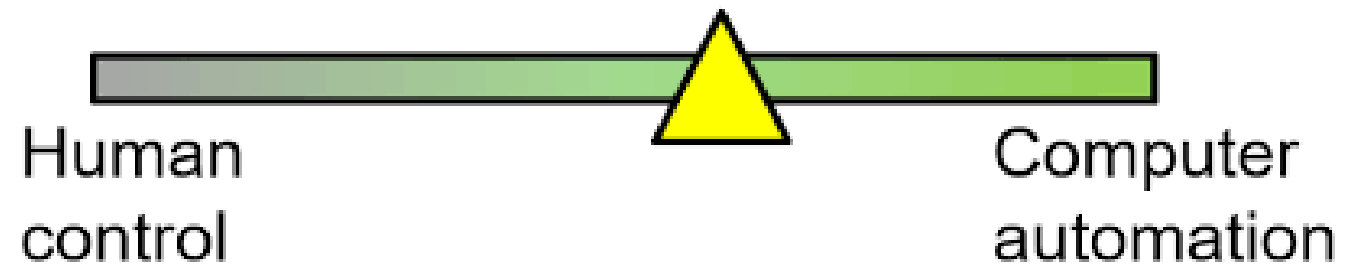


Human-centred Values in the Design of AI systems

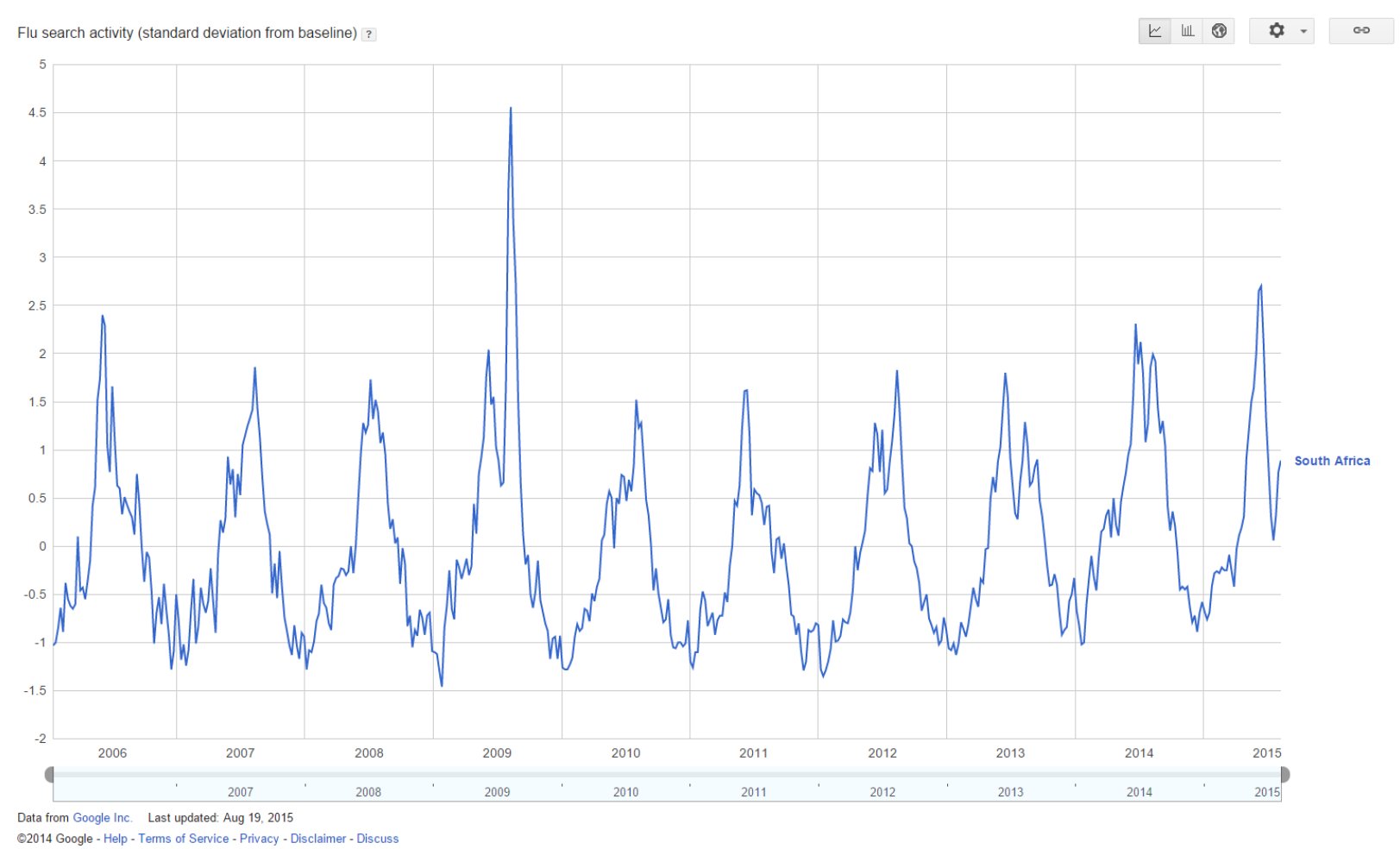
Automation, agency & fairness

Dr Georgia Panagiotidou

Automation & Agency



Google Flu Trends Failure



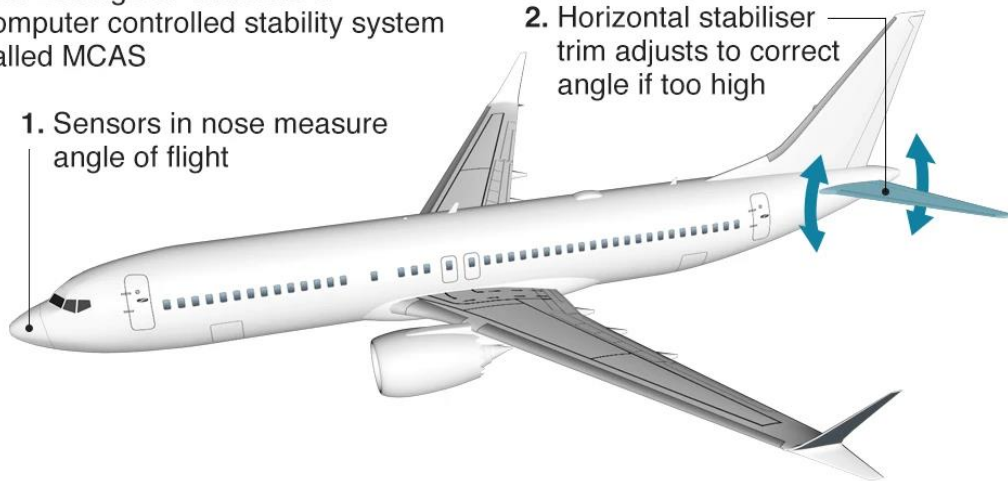
Boeing MCAS Failure

How the MCAS system works

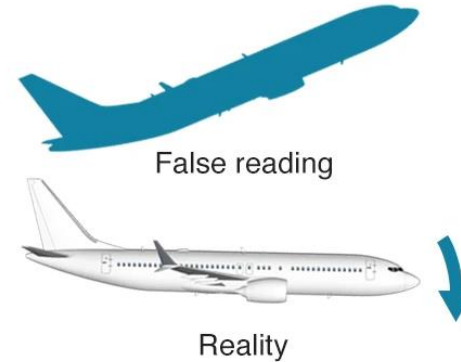
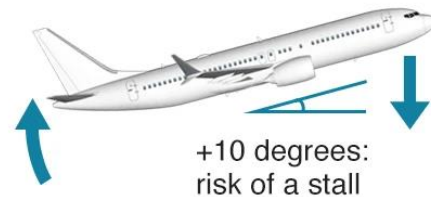
The Boeing 737 Max has a computer controlled stability system called MCAS

1. Sensors in nose measure angle of flight

2. Horizontal stabiliser trim adjusts to correct angle if too high



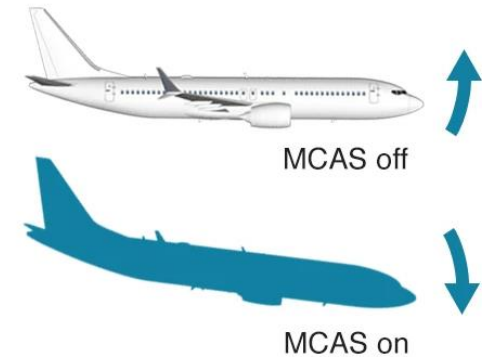
3. Nose pushed down to reduce risk of a stall



4. But if the sensor reading is wrong, MCAS may activate and push the nose down anyway

5. Pilots can temporarily switch off MCAS and pull up.

But system restarts if false readings continue, creating a tug of war between the aircraft and its crew

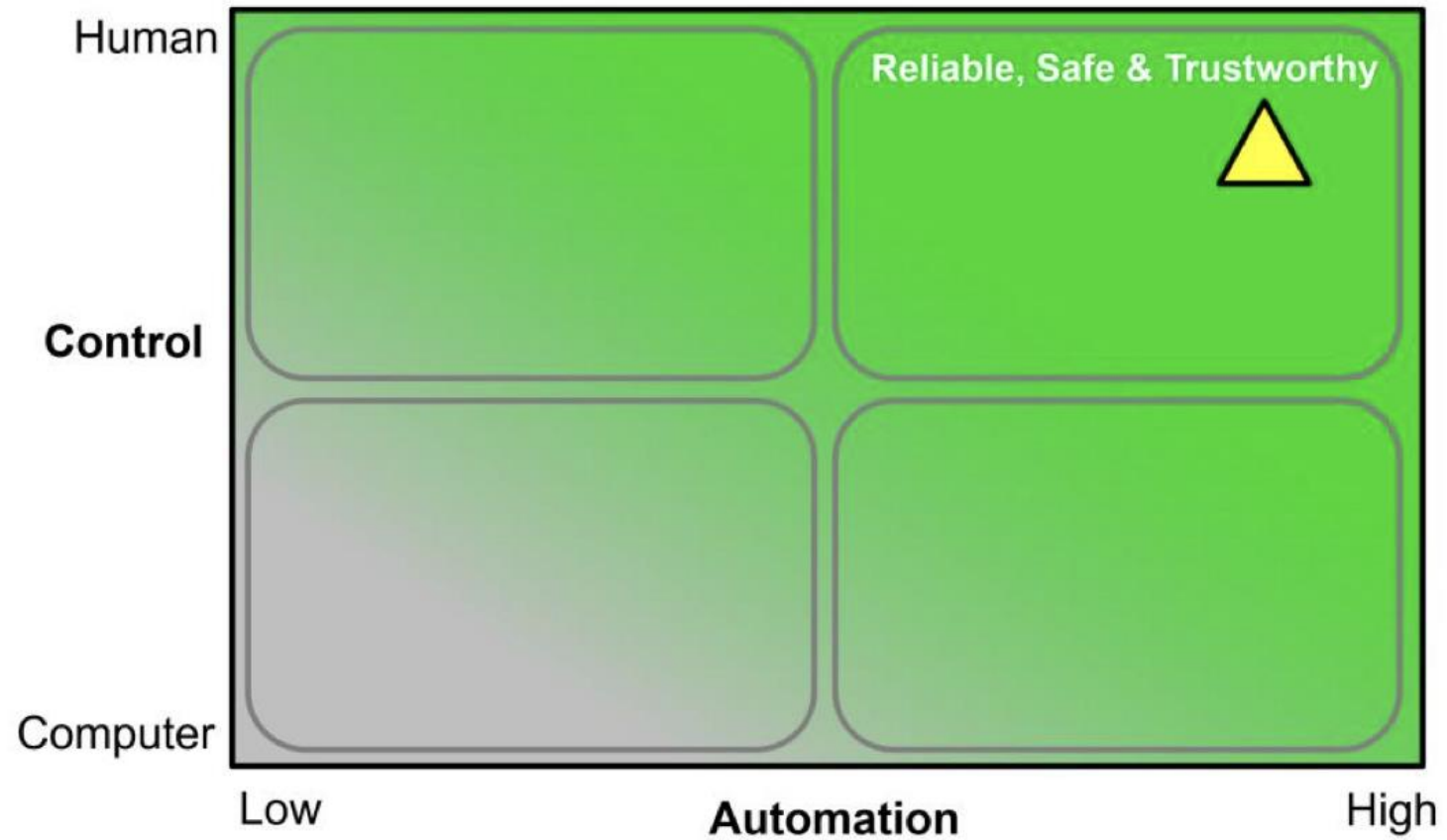


Source: Boeing, The Air Current

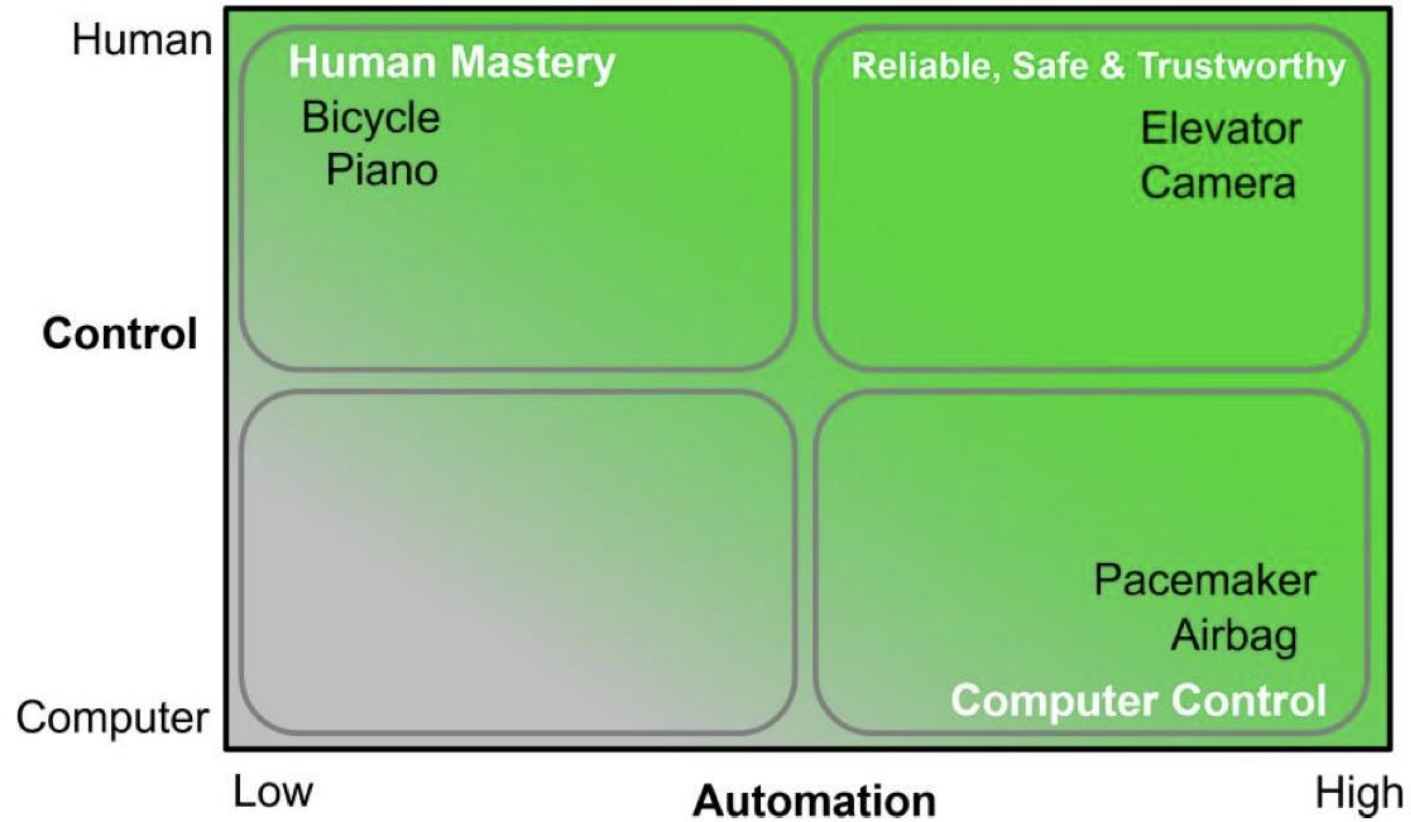
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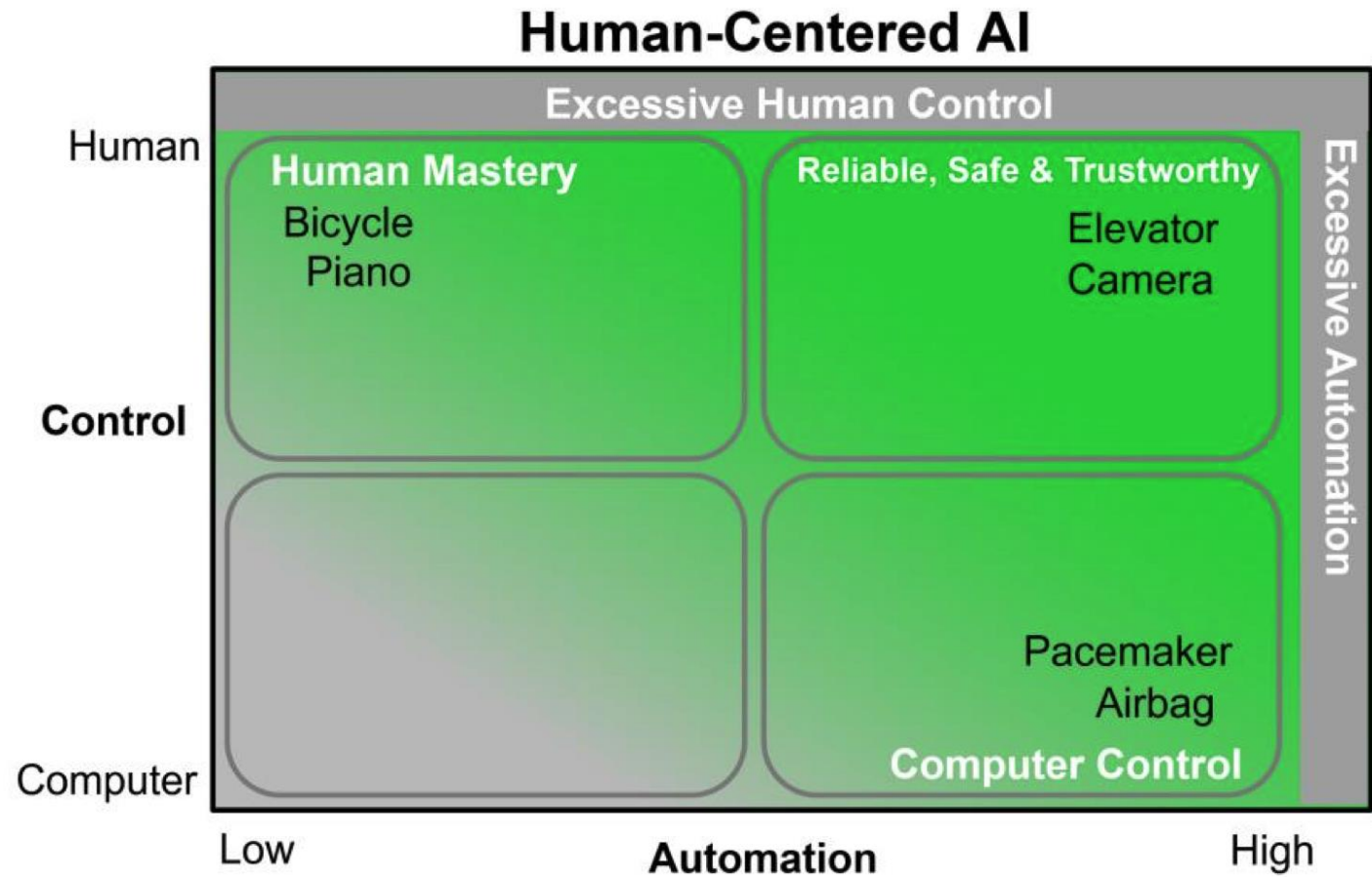
<https://www.bbc.co.uk/news/business-55751150>

Human-Centered AI



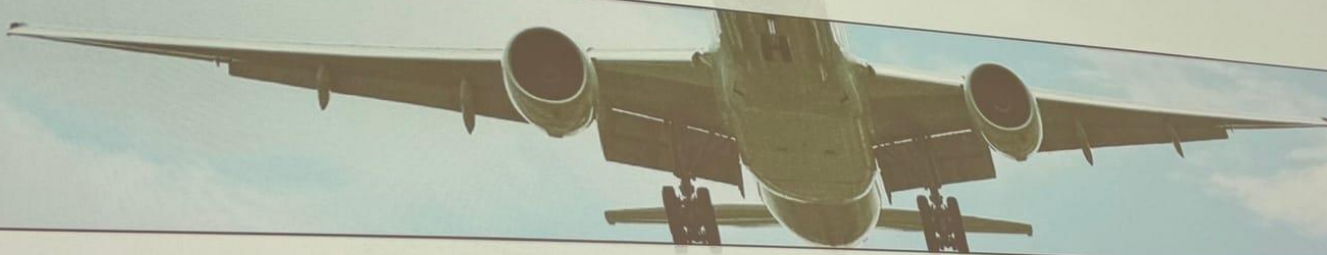
Human-Centered AI





Moral Crumple Zones:

What happens to “humans in the loop” who exist primarily to take the blame when an AI-human partnership goes wrong.



Elish, Madeleine Clare. “Moral Crumple Zones: Cautionary Tales in Human-Robot Interaction.” *Engaging Science, Technology, and Society* 5 (March 23, 2019): 40–60.

Automating work with GenAI tools

Han Qiao, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2025. To Use or Not to Use: Impatience and Overreliance When Using Generative AI Productivity Support Tools. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 1122, 1–18.



To Use or Not to Use: Impatience and Overreliance When Using Generative AI Productivity Support Tools

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Abstract

Generative AI has the potential to assist people with completing various tasks, but increased productivity is not guaranteed due to challenges such as uncertainty in output quality and unclear processing time. Through an online crowdsourced experiment (N=508), leveraging a “paint by numbers” task to simulate properties of GenAI assistance, we explore how, and how well, users make decisions on whether to use or not use automation to maximize their productivity given varying waiting times and output quality. We observed gaps between user’s actual choices and their optimal choices and characterized these gaps as the “gulf of impatience” and the “gulf of overreliance”. We also distilled strategies that participants adopted when making their decisions. We discuss design considerations in supporting users to make more informed decisions when interacting with GenAI tools and make these tools more useful for improving users’ task performance, productivity and satisfaction.

CCS Concepts

• **Human-centered computing** → **Laboratory experiments; Empirical studies in HCI.**

Keywords

generative AI, decision-making, productivity, reliance, AI, automation, controlled experiment

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Han Qiao, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2025. To Use or Not to Use: Impatience and Overreliance When Using Generative AI Productivity Support Tools. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*. April 26–May 01, 2025, Yokohama, Japan. ACM.

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Recent advancements in Generative AI (GenAI) have led to the development of a wide variety of GenAI support tools that assist people in completing various tasks across different domains (including writing, design, manufacturing, video production, education and programming). GenAI support tools have proven to effectively assist people with divergent thinking, especially associated with ideation and prototyping in early stages of design projects [31, 58, 69], as well as in end-to-end task completion with specifications and requirements, associated with tasks such as programming [11, 39], manufacturing [13] and landscape rendering [20]. However, despite their capabilities, GenAI tools require time to generate responses, often produce uncertain outputs, and demand additional effort from users to make further edits and correct mistakes [13, 32, 53, 61]. Therefore, the use of GenAI support tools does not always result in guaranteed improvements in productivity [39, 53].

For users seeking ideation support, or for novices who are not able to complete tasks without the assistance of GenAI, productivity may not be the primary concern. However, for users who are able to complete tasks like writing, 3D modeling and programming on their own but are now offered GenAI assistance, the uncertainty around the tools’ impact on productivity raises important questions about how individuals engage with these tools to reach their optimal task performance. Therefore, in this paper, we explore the following research questions:

- RQ1: Given the goal of maximizing productivity, how good are people at making optimal decisions about when to use GenAI?
- RQ2: How do people decide when to use or not use GenAI tools?

Automating work with GenAI tools

RQ1: Given the goal of maximizing productivity, how good are people at making optimal decisions about when to use GenAI?

RQ2: How do people decide when to use or not use GenAI tools?



To Use or Not to Use: Impatience and Overreliance When Using Generative AI Productivity Support Tools

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Research Methods

Manual or Assisted Fill

Latency condition: 0 - 210 secs

Error rate: 0-75%

508 participants

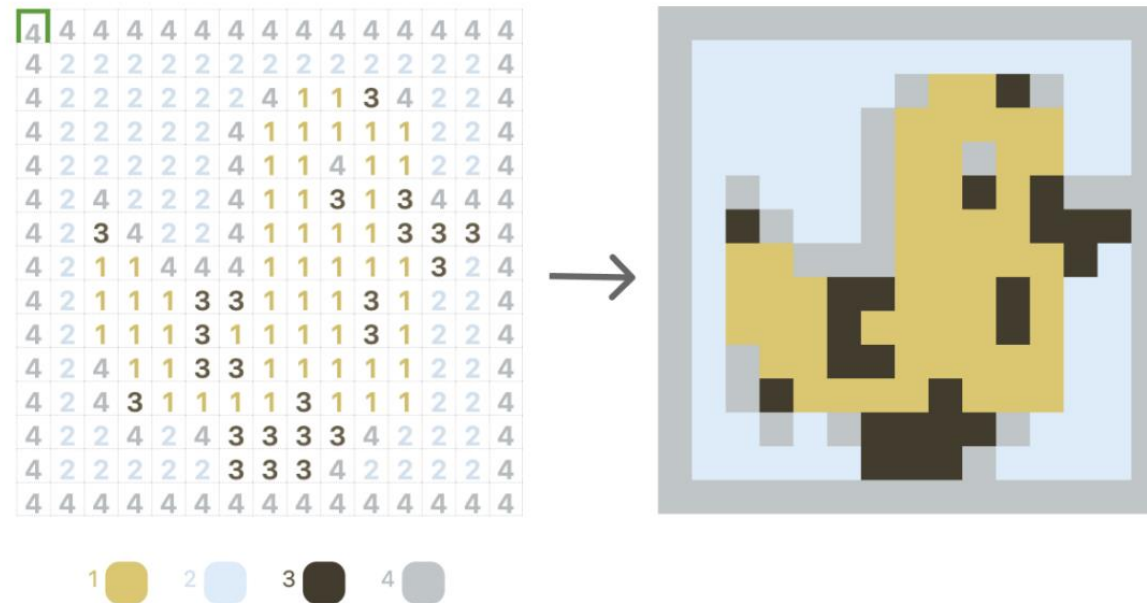
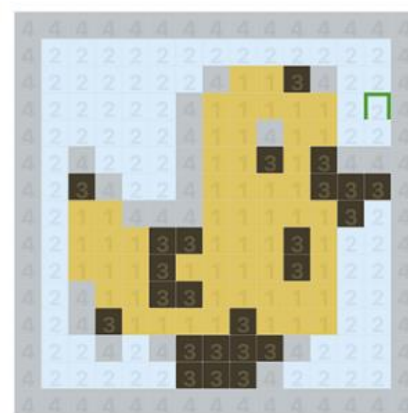
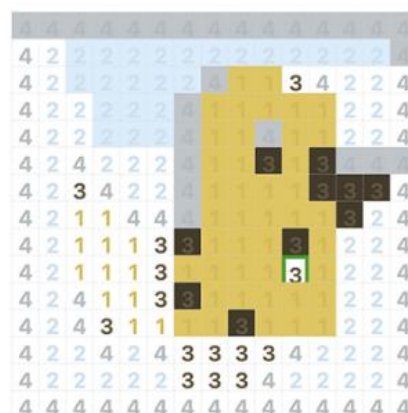
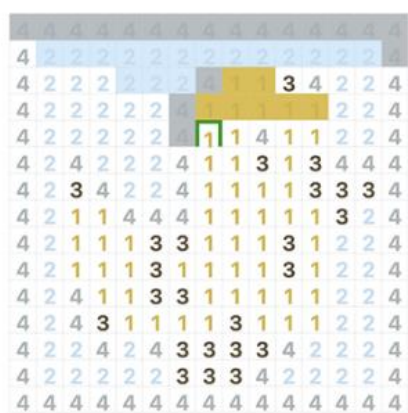
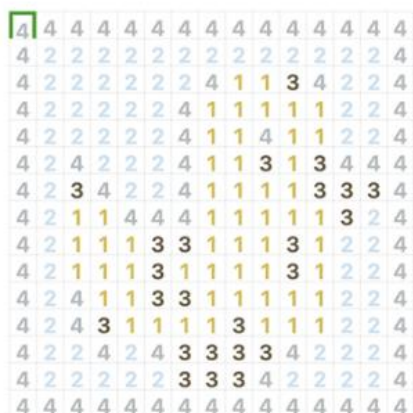


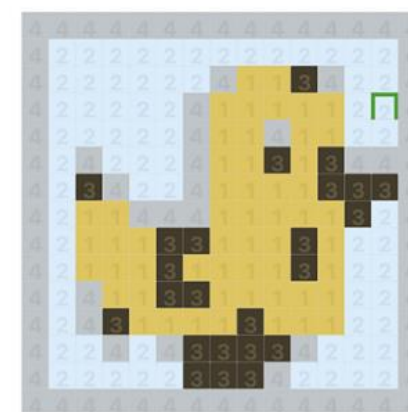
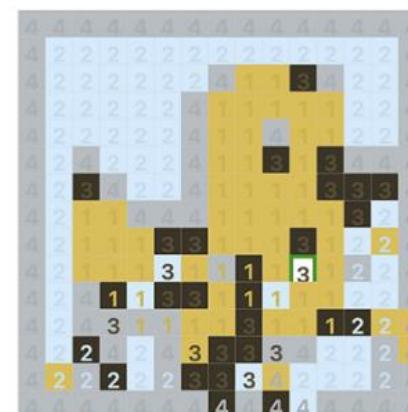
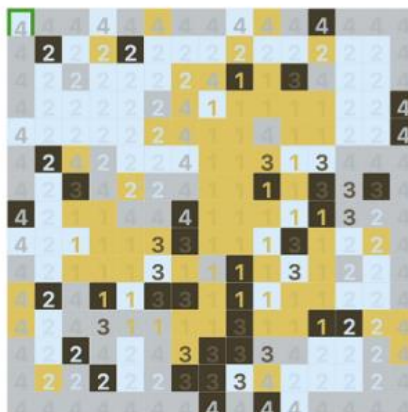
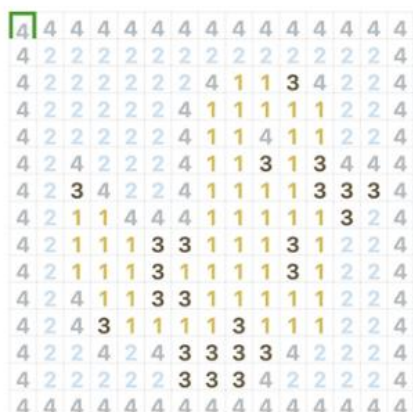
Figure 1: An example of an empty grid (left) and the final image presented to users (right) in a paint by numbers task.

Start Manual Fill



Manual Fill: Fill in colors by navigating using arrow keys and filling in colors using number keys.

Start Assisted Fill



Assisted Fill: Wait for automation to finish filling in colors and then correct any mistakes.

1 Introduction and Practice

2 Task 1: Manual Fill

Manual Fill Completion Time: 180s

Assisted Fill Tool Performance

Latency: 150s Error Rate: 35%

3 Task 2: Assisted Fill

Assisted Fill Completion Time: 200s
150s Latency + 50s

Question 1

For Assisted Fill tools, if the error rate is always 35%, but the wait time can vary between 0 second to 210 seconds:
What is the longest you would be willing to wait for the Assisted Fill Tool?

Remember, you are trying to complete the overall task in as little time as possible.

Question 2

For Assisted Fill tools, if the waiting time is always 150 seconds, but the error rate can vary between 0% to 75%:
What is the highest error rate you would be willing to use for the Assisted Fill Tool?

Remember, you are trying to complete the overall task in as little time as possible.

Assisted Fill

Latency: 60s

Error Rate: 35%

OR

Manual Fill

Question 3

Decision-Making Strategies

How did you decide how long you are willing to wait for Assisted Fill for task 3?

And how did you decide at what error rate you are still willing to use Assisted Fill for task 3?

What factors influenced your decision?

4 Pre-Task 3 Decision Making

Answer: Latency 45s

Wait time of your previous Assisted Fill was: 150s

Answer: Error Rate 10%

Error rate of your previous Assisted Fill was: 35%

5 Task 3

The latency of this 35% error rate Assisted Fill tool is 60s, which is greater than the participant's answer of 45s.

So the participant will need to finish Task 3 with Manual Fill.

6 Post-Task 3 Survey

Findings

- People make optimal decisions when Assisted Fill is either high performing in both latency and error rate or low performing in both latency and error rate. [“smart AI” and “dumb AI”.]
- The ***gulf of impatience*** occurs when the AI’s performance becomes more moderate.
- The ***gulf of overreliance*** occurs especially when one of the two conditions is extremely high performing, and the other is low performing [“smart but slow AI” or “dumb but fast AI”]

GenAI and Critical Thinking

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

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Abstract

The rise of Generative AI (GenAI) in knowledge workflows raises questions about its impact on critical thinking skills and practices. We survey 319 knowledge workers to investigate 1) when and how they perceive the enactment of critical thinking when using GenAI, and 2) when and why GenAI affects their effort to do so. Participants shared 936 first-hand examples of using GenAI in work tasks. Quantitatively, when considering both task- and user-specific factors, a user's task-specific self-confidence and confidence in GenAI are predictive of whether critical thinking is enacted and the effort of doing so in GenAI-assisted tasks. Specifically, higher confidence in GenAI is associated with less critical thinking, while higher self-confidence is associated with more critical thinking. Qualitatively, GenAI shifts the nature of critical thinking toward information verification, response integration, and task stewardship.

Confidence Effects From a Survey of Knowledge Workers. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 23 pages. <https://doi.org/10.1145/3706598.3713778>

1 Introduction

Generative AI (**GenAI**) tools, defined as any “end user tool [...] whose technical implementation includes a generative model based on deep learning”,¹ are the latest in a long line of technologies that raise questions about their impact on the quality of human thought, a line that includes writing (objected to by Socrates), printing (objected to by Trithemius), calculators (objected to by teachers of arithmetic), and the Internet.

Such consternation is not unfounded. Used improperly, technology

GenAI and Critical Thinking

RQ1 When and how do knowledge workers perceive the enactment of critical thinking when using GenAI?

RQ2 When and why do knowledge workers perceive increased/decreased effort for critical thinking due to GenAI?

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

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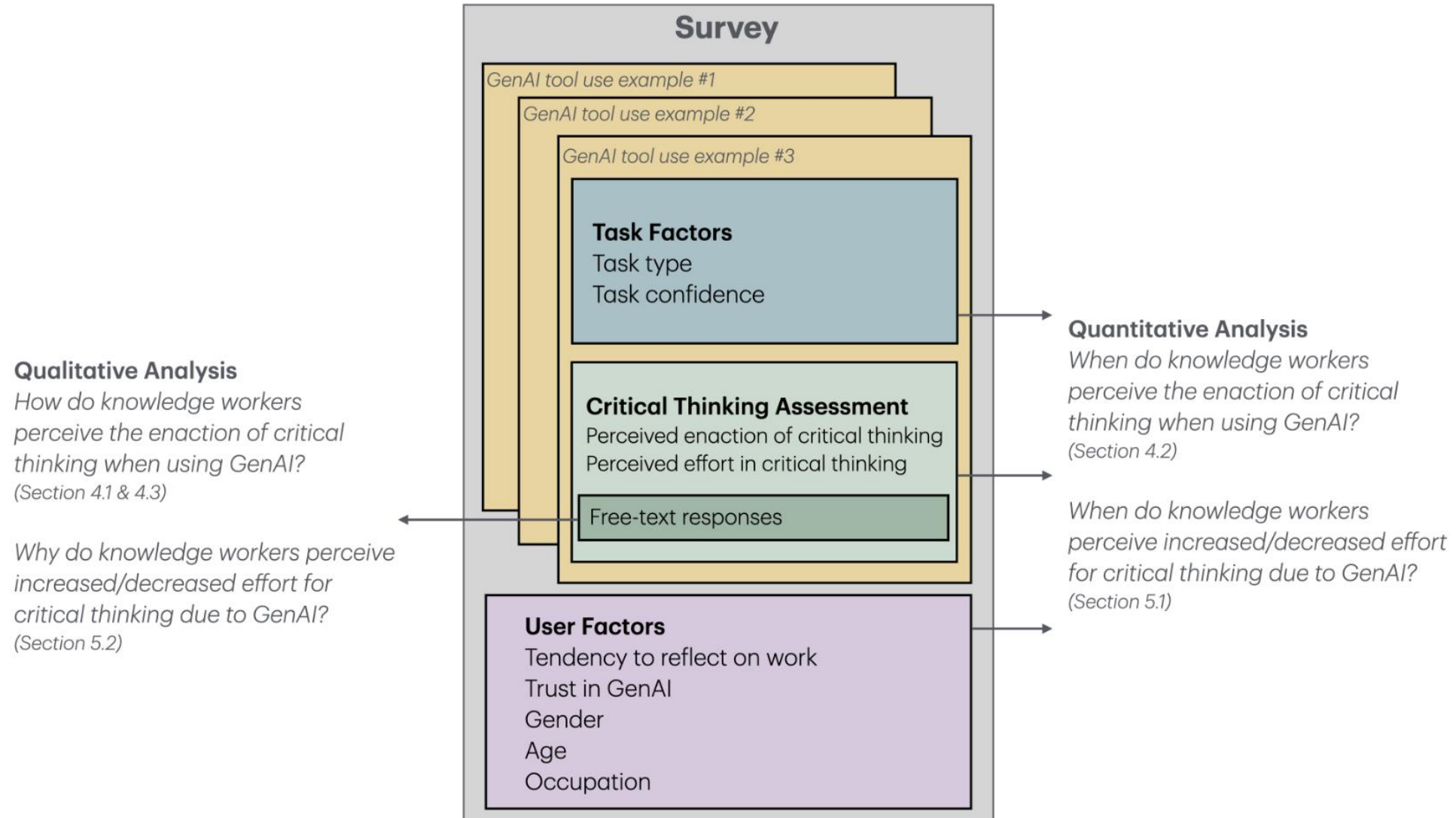
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Such consternation is not unfounded. Used improperly, technology can lead to a loss of critical thinking skills and a reliance on the

319 survey responses on Prolific



Findings

1. *a higher confidence in GenAI is associated with less critical thinking even though it is perceived as less effort to do so, and*
2. *a higher self-confidence is associated with more critical thinking even though it is perceived as more effort to do so.*

Findings

1. *Design strategies should focus on balancing these aspects*
 1. *AI tools could incorporate feedback mechanisms that help users gauge the reliability of AI outputs, when to trust the AI and when to apply their critical thinking skills*
 2. *Enhance awareness of critical thinking opportunities: indicate that knowledge workers tend to forgo critical thinking for tasks perceived as unimportant or secondary.*

Human-centred Values in the Design of AI systems

Automation, agency & fairness

Dr Georgia Panagiotidou

Fairness

Fairness is the absence of any prejudice or favouritism toward an individual or group based on their inherent or acquired characteristics.

Unfair Systems that Demonstrate Discrimination

Two Drug Possession Arrests

 DYLAN FUGETT	 BERNARD PARKER
LOW RISK 3	HIGH RISK 10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Two Drug Possession Arrests

DYLAN FUGETT	BERNARD PARKER
Prior Offense 1 attempted burglary	Prior Offense 1 resisting arrest without violence
Subsequent Offenses 3 drug possessions	Subsequent Offenses None
LOW RISK 3	HIGH RISK 10

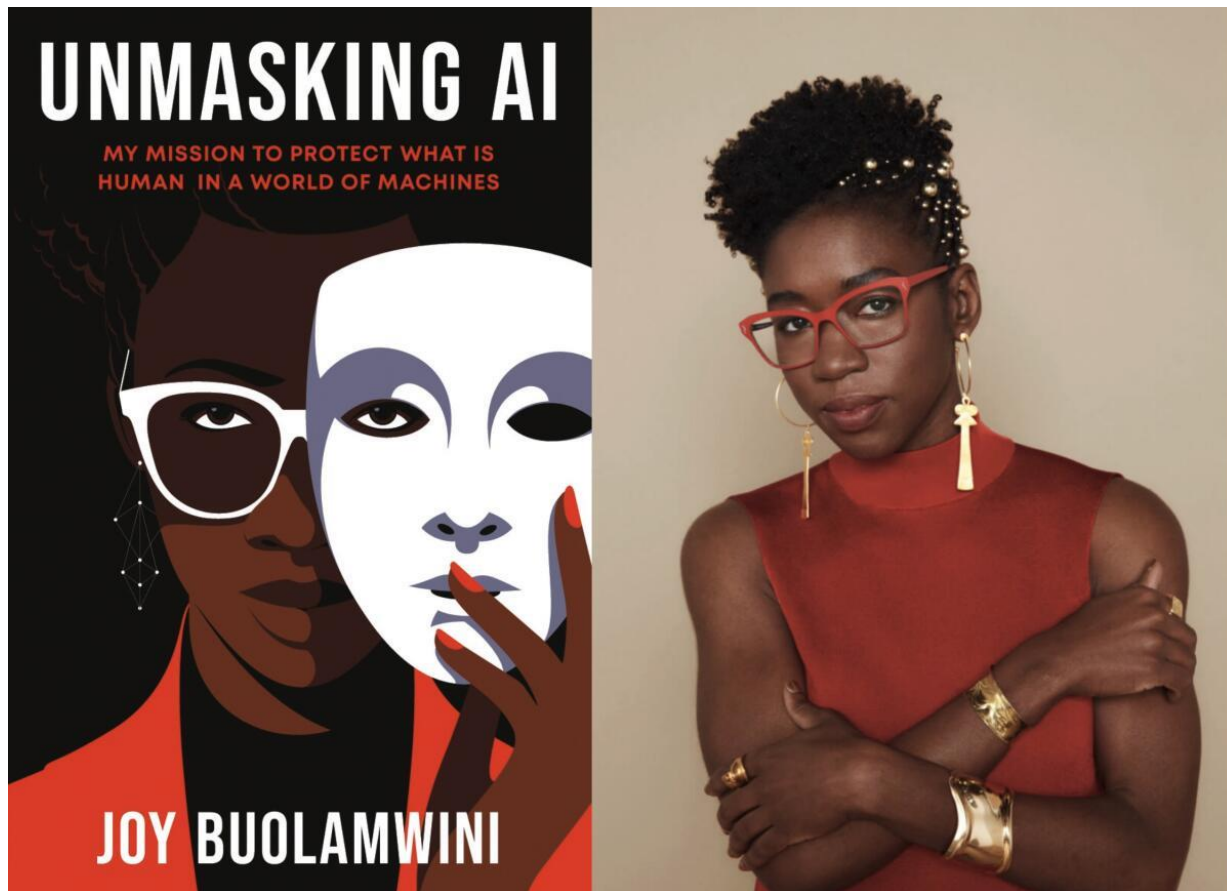
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<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>



<https://time.com/archive/6906847/are-face-detection-cameras-racist/>

Algorithmic Bias - “The coded gaze”

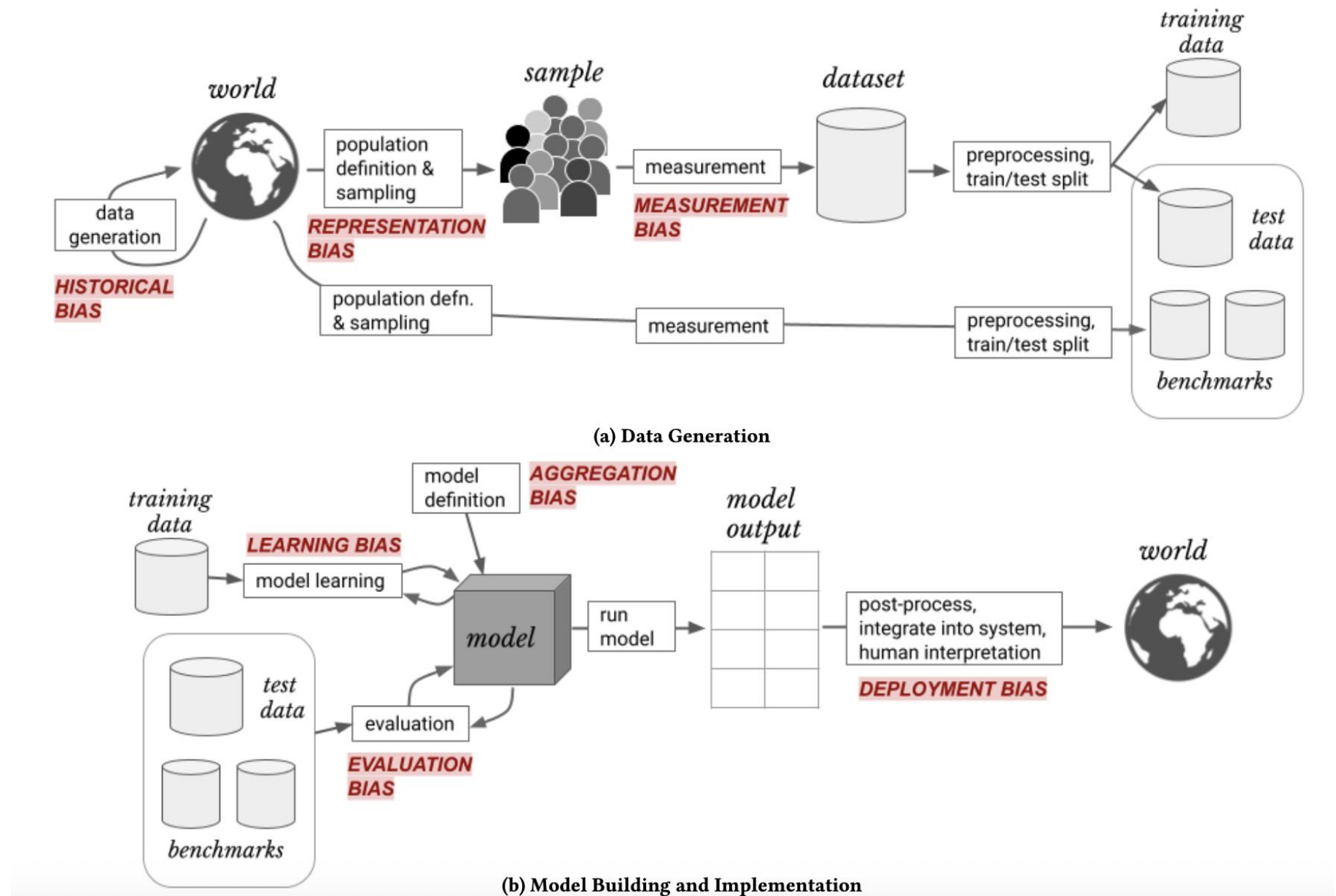


Joy Buolamwini, 2023

The harms of information systems

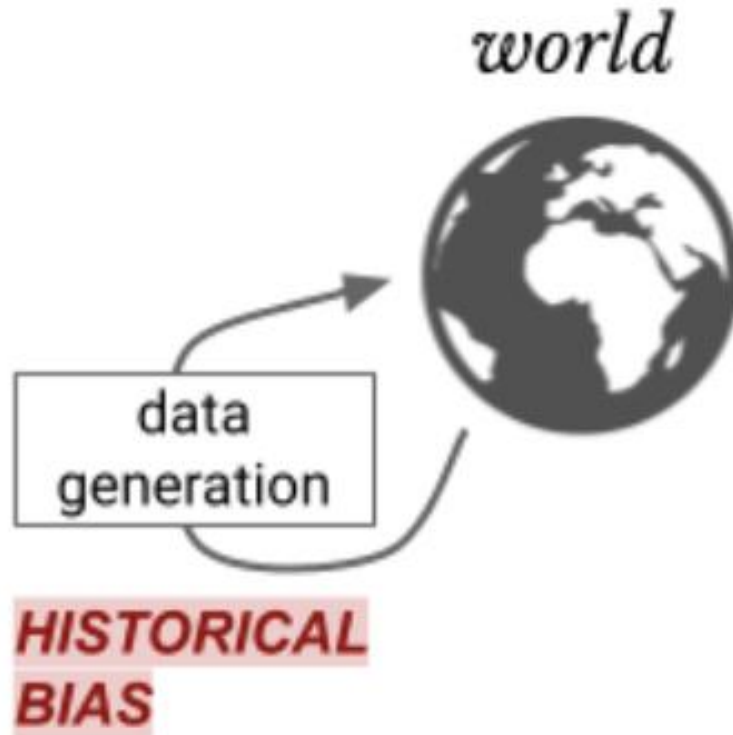
- Allocative Harms
 - caused when a system withholds certain groups an opportunity or a resource.
- Representational Harms
 - occur when systems reinforce the subordination of some groups along the lines of identity—race, class, gender, etc.

Sources of Harm throughout the Machine Learning Life Cycle



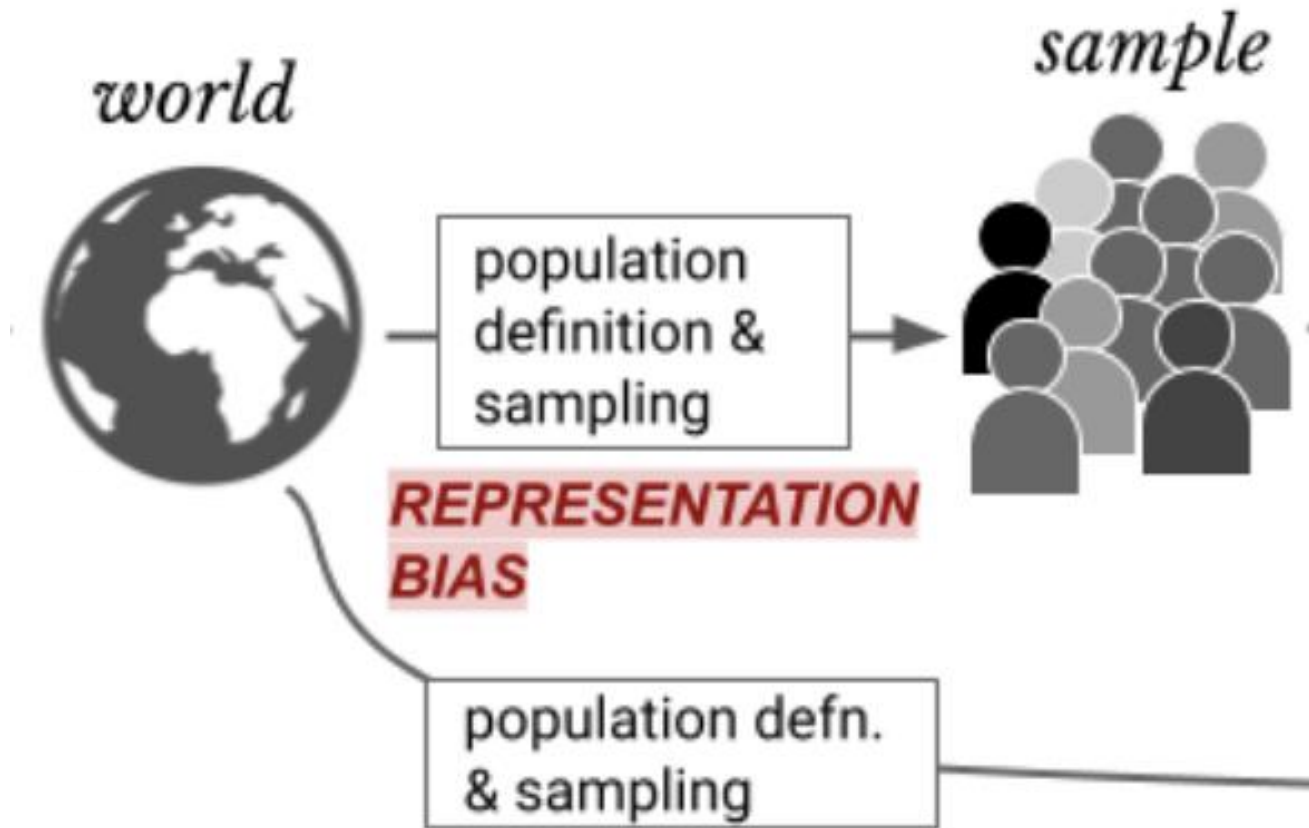
Historical Bias

e.g. Word embeddings



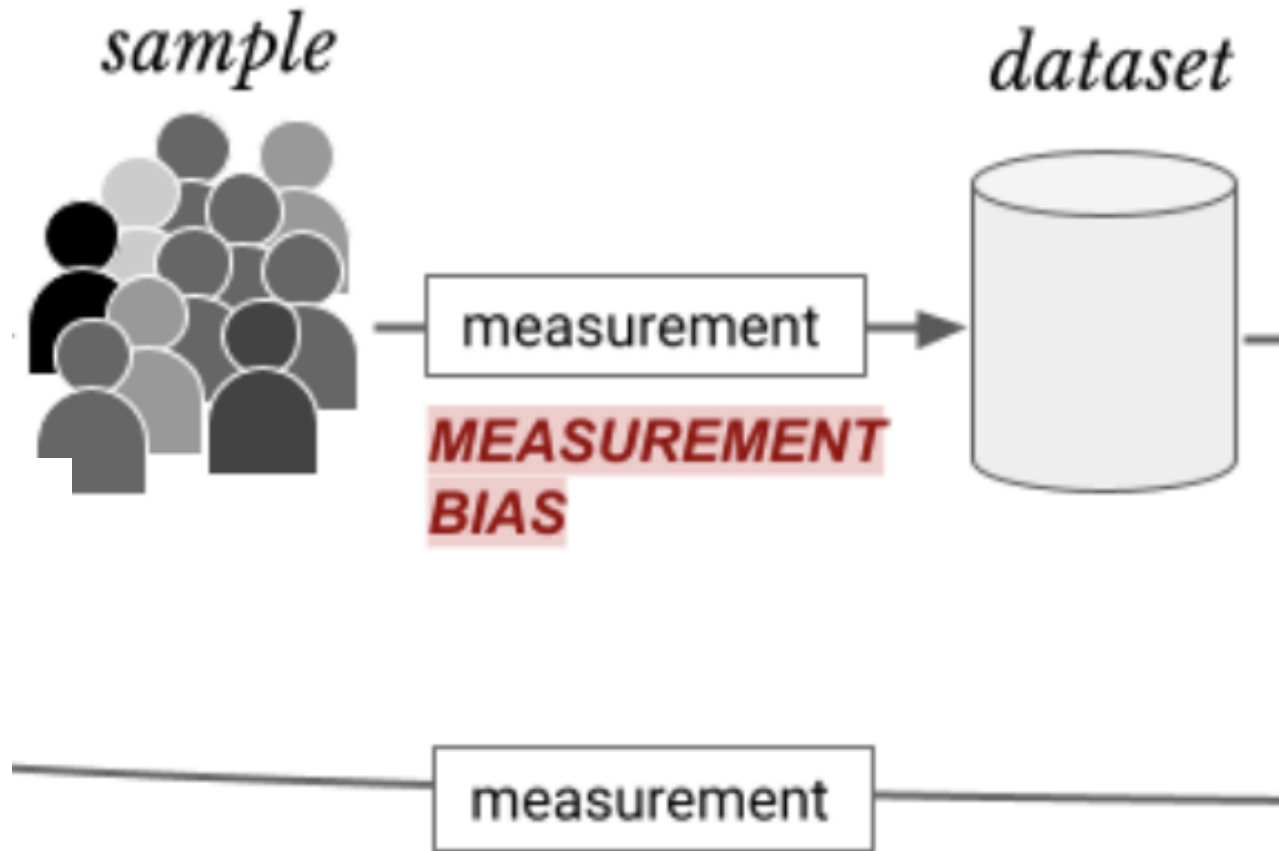
Representation Bias

e.g. ImageNet



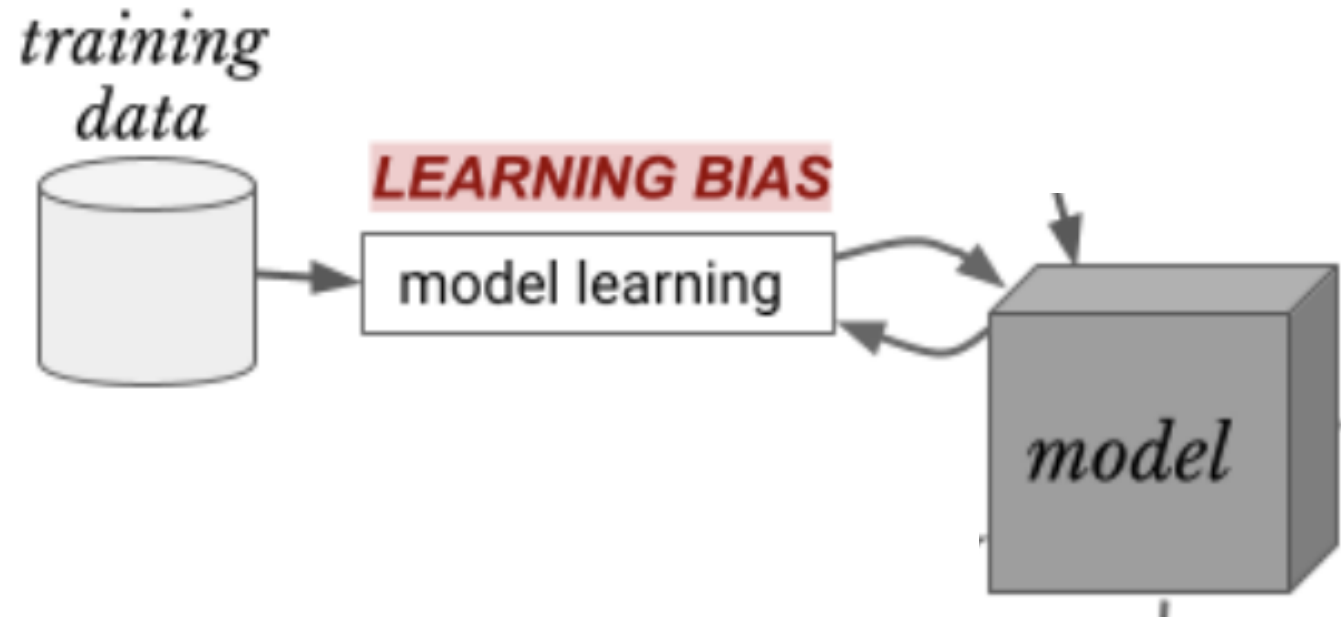
Measurement Bias

e.g. COMPAS



Learning Bias

e.g. optimising for compactness



Transparency & Explainability

- **Transparency** means AI systems should not operate as "black boxes." Instead, users should have insights into the AI's reasoning process.
- **Explainability** refers to the ability of AI systems to justify their outputs in a way that humans can understand.

Step 1: Run a tokenized source prompt, `United Kingdom`, through the source model. Each transformer layer is denoted with ℓ^i for clarity.

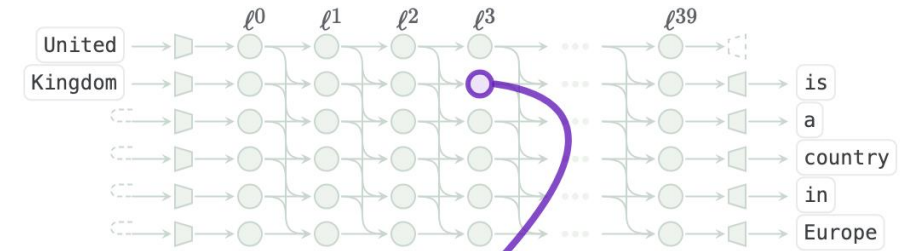
Step 2: Extract the hidden representation of interest from the source location (i.e., a token position at a layer in the model). We explain one method for selecting a source location in the case studies that follow.

Step 3: Run the inspection prompt, `The largest city in x`, through the target model, stopping at a pre-determined target location. In the case studies that follow, we discuss two strategies to select a target location to answer specific questions about a given source location.

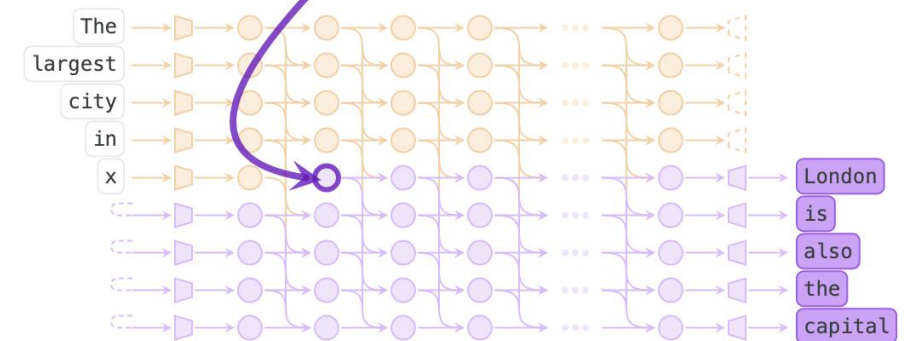
Step 4: At the target location, inject the hidden representation extracted in Step 2. We call this the "patching operation".

Step 5: Continue generation for the inspection prompt, with the patched hidden representation taking the place of the `x` token

Source Prompt



Inspection Prompt



<https://pair.withgoogle.com/explorables/patchscopes/>

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